

KisanGat

Implementation Plan

Version 1.0 MVP

Estimated Duration: 8–12 Weeks

Development Methodology: Agile Sprint-Based Development

Phase 0: Project Planning & Architecture

Duration: 2–3 Days

Goal: Prepare project foundations before coding.

Tasks:

- Finalize PRD
- Finalize App Flow
- Finalize Database Schema
- Finalize Technical Requirements
- Finalize UI/UX Design System
- Define folder structure
- Create Git repository
- Create development workflow

Deliverables:

✓ PRD

✓ TRD

✓ App Flow Document

✓ Database Schema

✓ UI/UX Design Brief

✓ Git Repository

✓ Sprint Plan

Phase 1: Environment & Project Setup

Duration: 2-3 Days

Goal: Create frontend and backend foundations.

Backend Setup

Create Django Project

Install:

```
Django
Django REST Framework
SimpleJWT
django-cors-headers
mysqlclient
python-dotenv
```

Setup:

```
backend/
```

Configure:

- MySQL Connection
 - Environment Variables
 - DRF Configuration
 - JWT Configuration
-

Frontend Setup

Create React Application

Install:

```
React
Redux Toolkit
React Router
Axios
Material UI
Recharts
```

React Hook Form
Zod

Setup:

frontend/

Create:

- Theme Provider
- Routing
- Axios Client
- Layout System

Deliverables

- ✓ Backend Boilerplate
- ✓ Frontend Boilerplate
- ✓ Environment Setup
- ✓ Docker Skeleton

Phase 2: Authentication Module

Duration: 3 Days

Goal: Secure Admin Access

Backend

Create:

users

refresh_tokens

Develop:

POST /auth/login

POST /auth/refresh

POST /auth/logout

POST /auth/change-password

Implement:

- JWT Authentication
 - Password Hashing
 - Token Rotation
-

Frontend

Screens:

Login

Change Password

Features:

- Validation
 - Protected Routes
 - Session Handling
-

Deliverables

✓ Login System

✓ JWT Authentication

✓ Protected Routes

✓ Session Persistence

Phase 3: Core Layout & Dashboard

Duration: 4 Days

Goal: Build application shell.

Frontend

Create:

Sidebar

Topbar

Breadcrumbs

Notifications

Dashboard Layout

Dashboard Components

KPI Cards

Charts

Tables

Summary Widgets

Backend

Dashboard APIs

GET /dashboard/summary

GET /dashboard/revenue

GET /dashboard/inventory

Deliverables

✓ Dashboard

✓ Sidebar Navigation

✓ Top Navigation

✓ Responsive Layout

Phase 4: Farmer Management

Duration: 4 Days

Goal: Manage farmer records.

Backend

Tables:

farmers

farmer_ledger

Develop APIs:

GET Farmers

Create Farmer

Update Farmer

Delete Farmer

Farmer Details

Farmer Ledger

Frontend

Pages:

Farmer List

Add Farmer

Edit Farmer

Farmer Detail

Ledger View

Deliverables

✓ Farmer CRUD

✓ Farmer Detail Screen

✓ Farmer Ledger

Phase 5: Milk Collection Module

Duration: 5 Days

Goal: Record milk collection.

Backend

Tables:

milk_collections

milk_rates

inventory_transactions

Features:

- Auto Amount Calculation
- Rate Snapshot
- Inventory Update

APIs:

Collection CRUD

Collection Reports

Frontend

Pages:

Collection List

Add Collection

Edit Collection

Collection Detail

Deliverables

✓ Milk Collection System

✓ Inventory Integration

✓ Collection Reports

Phase 6: Customer Management

Duration: 4 Days

Goal: Manage customers.

Backend

Tables:

customers

customer_ledger

APIs:

Customer CRUD

Customer Detail

Customer Ledger

Frontend

Pages:

Customer List

Add Customer

Customer Detail

Ledger

Deliverables

✓ Customer Management

✓ Customer Ledger

Phase 7: Milk Sales & Due Management

Duration: 6 Days

Goal: Manage milk sales.

Backend

Tables:

milk_sales

customer_payments

Features:

- Due Calculation
- Inventory Deduction
- Ledger Updates

APIs:

Sales CRUD

Payment Recording

Due Tracking

Frontend

Pages:

Sales List

Add Sale

Due Management

Payment Modal

Deliverables

✓ Sales System

✓ Due Management

✓ Payment Tracking

✓ Inventory Sync

Phase 8: Inventory Module

Duration: 3 Days

Goal: Track stock accurately.

Backend

Tables:

milk_inventory

inventory_transactions

Features:

- Opening Stock
 - Closing Stock
 - Carry Forward
 - Adjustments
-

Frontend

Pages:

Inventory Dashboard

Inventory History

Inventory Adjustment

Deliverables

✓ Inventory Management

✓ Stock Reconciliation

✓ Adjustment System

Phase 9: Farmer Payments

Duration: 4 Days

Goal: Automate settlements.

Backend

Tables:

farmer_payments

Features:

- Daily Payments
 - Weekly Payments
 - Monthly Payments
-

Frontend

Pages:

Payment List

Generate Payment

Payment Preview

Receipt Download

Deliverables

✓ Payment Engine

✓ Settlement System

✓ Receipt Generation

Phase 10: Dairy Products

Duration: 4 Days

Goal: Manage dairy products.

Backend

Tables:

products

product_sales

product_inventory_transactions

Frontend

Pages:

Product List

Product Sales

Product Inventory

Deliverables

✓ Product Management

✓ Product Sales

✓ Product Inventory

Phase 11: Employee Management

Duration: 3 Days

Goal: Track employee records.

Backend

Tables:

employees

employee_attendance

employee_salary_records

Frontend

Pages:

Employee List

Attendance

Salary Records

Deliverables

✓ Employee Records

✓ Attendance Tracking

✓ Salary Tracking

Phase 12: Expense Management

Duration: 2 Days

Goal: Track expenses.

Backend

Tables:

expenses

APIs:

Expense CRUD

Expense Reports

Frontend

Pages:

Expense List

Add Expense

Expense Analytics

Deliverables

✓ Expense Tracking

✓ Expense Reports

Phase 13: Billing Engine

Duration: 5 Days

Goal: Generate PDF bills.

Backend

Tables:

generated_bills

farmer_bill_items

customer_bill_items

Libraries:

ReportLab

Features

Farmer Bill

Customer Bill

Bill Preview

PDF Download

Print Support

Frontend

Pages:

Bills

Bill Preview

Download

Print

Deliverables

✓ PDF Billing System

✓ Bill History

✓ Print Support

Phase 14: Reporting & Analytics

Duration: 5 Days

Goal: Business intelligence.

Backend

Reports:

Revenue

Collection

Inventory

Expenses

Dues

Farmer Payments

Product Sales

Frontend

Pages:

Reports Dashboard

Filters

Charts

Exports

Deliverables

✓ Analytics Module

✓ Reports Module

✓ Export Features

Phase 15: Settings Module

Duration: 2 Days

Goal: System configuration.

Backend

Tables:

dairy_settings

system_settings

Frontend

Pages:

Dairy Profile

Milk Rates

Admin Profile

Preferences

Deliverables

✓ Settings Module

✓ Rate Management

✓ Profile Management

Phase 16: Integration & Data Validation

Duration: 4 Days

Goal: Connect all modules.

Tasks

Inventory ↔ Sales

Inventory ↔ Collection

Ledger ↔ Payments

Rates ↔ Collection

Rates ↔ Sales

Reports ↔ Transactions

Bills ↔ Ledger

Deliverables

✓ Fully Connected System

✓ Business Logic Validation

Phase 17: Testing Phase

Duration: 1 Week

Goal: Ensure stability.

Backend Testing

Unit Tests

API Tests

Permission Tests

Validation Tests

Frontend Testing

Form Validation

Responsive Testing

Navigation Testing

Accessibility Testing

Deliverables

✓ Test Reports

✓ Bug Fixes

✓ Stable Build

Phase 18: Docker & Deployment

Duration: 3 Days

Goal: Production readiness.

Create:

Dockerfile (Frontend)

Dockerfile (Backend)

docker-compose.yml

Nginx Config

Production Stack

Internet

↓

Nginx

↓

React Build

↓

Gunicorn

↓

Django API

↓

MySQL

Deliverables

✓ Docker Setup

✓ Production Build

✓ Nginx Configuration

✓ SSL Ready

Phase 19: Final Polish

Duration: 3 Days

Goal: Production quality UX.

Tasks

Loading States

Empty States

Error States

Success Toasts

Accessibility Review

Performance Optimization

Code Cleanup

Documentation

Deliverables

✓ Production Ready Application

✓ Technical Documentation

✓ API Documentation

✓ Deployment Guide

✓ User Manual

Final Deliverables

Frontend

- React Application
- Responsive UI
- Dashboard
- Reports

Backend

- Django REST APIs
- JWT Authentication
- Business Logic

Database

- MySQL Schema
- Migrations
- Seed Data

Infrastructure

- Docker
- Nginx
- Deployment Scripts

Documentation

- PRD
- TRD
- App Flow
- Database Schema
- API Documentation
- Deployment Guide

Estimated MVP Completion: 8–12 Weeks (Single Developer) 4–6 Weeks (2 Developers) 2–3 Weeks (AI-Assisted Development Team)